

01_deduplication.R

```
packages <- c("readr", "readxl", "dplyr", "stringr", "janitor", "writexl")

installed_packages <- rownames(installed.packages())

for (p in packages) {
  if (!(p %in% installed_packages)) {
    install.packages(p)
  }
}

library(readr)
library(readxl)
library(dplyr)
library(stringr)
library(janitor)
library(writexl)

folder <- "."

scopus_file <- file.path(folder, "scopus.csv")
wos_file <- file.path(folder, "wos.xlsx")

if (!file.exists(scopus_file)) {
  stop("Nije pronađena Scopus CSV datoteka u zadanom direktoriju.")
}

if (!file.exists(wos_file)) {
  stop("Nije pronađena WoS XLS/XLSX datoteka u zadanom direktoriju.")
}

cat("Pronađena Scopus datoteka:\n")
cat(scopus_file, "\n\n")

cat("Pronađena WoS datoteka:\n")
cat(wos_file, "\n\n")

scopus <- read_csv(scopus_file, show_col_types = FALSE) %>%
  clean_names() %>%
  mutate(database = "Scopus")

wos <- read_excel(wos_file) %>%
  clean_names() %>%
  mutate(database = "WoS")

cat("Nazivi stupaca u Scopus datoteci:\n")
print(names(scopus))

cat("\nNazivi stupaca u WoS datoteci:\n")
```

```

print(names(wos))

find_col <- function(data, possible_names) {
  existing <- possible_names[possible_names %in% names(data)]
  if (length(existing) == 0) {
    return(NA_character_)
  } else {
    return(existing[1])
  }
}

scopus_title_col <- find_col(
  scopus,
  c("title", "article_title", "document_title")
)

scopus_doi_col <- find_col(
  scopus,
  c("doi")
)

scopus_year_col <- find_col(
  scopus,
  c("year", "publication_year")
)

wos_title_col <- find_col(
  wos,
  c("article_title", "title", "ti", "document_title")
)

wos_doi_col <- find_col(
  wos,
  c("doi", "di")
)

wos_year_col <- find_col(
  wos,
  c("publication_year", "year", "py")
)

cat("\nPrepoznati stupci:\n")
cat("Scopus naslov:", scopus_title_col, "\n")
cat("Scopus DOI:", scopus_doi_col, "\n")
cat("Scopus godina:", scopus_year_col, "\n")
cat("WoS naslov:", wos_title_col, "\n")
cat("WoS DOI:", wos_doi_col, "\n")
cat("WoS godina:", wos_year_col, "\n\n")

```

```
if (is.na(scopus_title_col)) {  
  stop("Nije pronađen stupac s naslovom u Scopus datoteci.")  
}
```

```
if (is.na(wos_title_col)) {  
  stop("Nije pronađen stupac s naslovom u WoS datoteci.")  
}
```

```
scopus2 <- scopus %>%  
  mutate(  
    title_clean = str_to_lower(  
      str_squish(  
        as.character(.data[[scopus_title_col]])  
      )  
    ),  
    doi_clean = if (!is.na(scopus_doi_col)) {  
      str_to_lower(  
        str_squish(  
          as.character(.data[[scopus_doi_col]])  
        )  
      )  
    } else {  
      NA_character_  
    },  
    year_clean = if (!is.na(scopus_year_col)) {  
      as.character(.data[[scopus_year_col]])  
    } else {  
      NA_character_  
    }  
  )
```

```
wos2 <- wos %>%  
  mutate(  
    title_clean = str_to_lower(  
      str_squish(  
        as.character(.data[[wos_title_col]])  
      )  
    ),  
    doi_clean = if (!is.na(wos_doi_col)) {  
      str_to_lower(  
        str_squish(  
          as.character(.data[[wos_doi_col]])  
        )  
      )  
    } else {  
      NA_character_  
    },  
    year_clean = if (!is.na(wos_year_col)) {  
      as.character(.data[[wos_year_col]])  
    }
```

```

    } else {
      NA_character_
    }
  )
}

```

```
combined <- bind_rows(scopus2, wos2)
```

```
combined <- combined %>%
  mutate(
    doi_clean = na_if(doi_clean, ""),
    title_clean = na_if(title_clean, ""),
    year_clean = na_if(year_clean, ""),
    duplicate_key = ifelse(
      !is.na(doi_clean),
      paste0("doi_", doi_clean),
      paste0("title_", title_clean, "_", year_clean)
    )
  )

```

```
unique_records <- combined %>%
  distinct(duplicate_key, .keep_all = TRUE)
```

```
duplicates <- combined %>%
  group_by(duplicate_key) %>%
  filter(n() > 1) %>%
  arrange(duplicate_key, database) %>%
  ungroup()

```

```
cat("\n===== STATISTIKA =====\n")
cat("Ukupno Scopus zapisa:", nrow(scopus), "\n")
cat("Ukupno WoS zapisa:", nrow(wos), "\n")
cat("Ukupno prije uklanjanja duplikata:", nrow(combined), "\n")
cat("Broj zapisa bez duplikata:", nrow(unique_records), "\n")
cat("Broj redaka označenih kao duplikati:", nrow(duplicates), "\n")
cat("Broj uklonjenih duplikata:", nrow(combined) - nrow(unique_records), "\n")
cat("=====\n")

```

```
write_xlsx(
  list(
    unique_records = unique_records,
    duplicates = duplicates,
    all_records = combined
  ),
  file.path(folder, "scopus_wos_deduplicated.xlsx")
)

```

```
write_csv(
  unique_records,
  file.path(folder, "scopus_wos_deduplicated.csv")
)

```

)

```
cat("\nGotovo!\n")
```

```
cat("Rezultati su spremljeni u:\n")
```

```
cat(folder, "\n\n")
```

```
cat("Izlazne datoteke:\n")
```

```
cat("1. scopus_wos_deduplicated.xlsx\n")
```

05_global_citations.R

```
library(readr)
library(dplyr)
library(stringr)
library(ggplot2)
library(forcats)
library(scales)

M <- read_csv("scopus_wos_deduplicated.csv", show_col_types = FALSE)

required_cols <- c("title", "year", "source_title", "cited_by")
missing_cols <- required_cols[!required_cols %in% names(M)]

if (length(missing_cols) > 0) {
  cat("Missing required columns:\n")
  print(missing_cols)
  stop("Some required columns are missing. Check names(M).")
}

has_authors <- "authors" %in% names(M)
has_doi <- "doi" %in% names(M)

current_year <- 2026

cited_docs <- M %>%
  transmute(
    title = str_squish(as.character(title)),
    authors = if (has_authors) str_squish(as.character(authors)) else NA_character_,
    year = as.numeric(year),
    source_title = str_squish(as.character(source_title)),
    doi = if (has_doi) str_to_lower(str_squish(as.character(doi))) else NA_character_,
    global_citations = as.numeric(cited_by)
  ) %>%
  mutate(
    global_citations = ifelse(is.na(global_citations), 0, global_citations),

    years_since_publication = pmax(current_year - year + 1, 1),
    citations_per_year = global_citations / years_since_publication,

    first_author = case_when(
      is.na(authors) | authors == "" ~ "Unknown",
      str_detect(authors, ";") ~ str_squish(str_split_fixed(authors, ";", 2)[, 1]),
      str_detect(authors, ",") ~ str_squish(str_split_fixed(authors, ",", 2)[, 1]),
      TRUE ~ str_squish(str_split_fixed(authors, " ", 2)[, 1])
    ),

    short_reference = paste0(first_author, " (", year, ")")
  ) %>%
  filter(
```

```

!is.na(title),
title != "",
!is.na(year)
)

write_csv(
  cited_docs,
  "documents_with_global_citations_and_citations_per_year.csv"
)

top_15_global_cited <- cited_docs %>%
  arrange(desc(global_citations), desc(citations_per_year)) %>%
  slice_head(n = 15) %>%
  mutate(
    rank_global = row_number(),
    citations_per_year = round(citations_per_year, 2)
  ) %>%
  select(
    rank_global,
    short_reference,
    year,
    title,
    source_title,
    global_citations,
    citations_per_year,
    doi
  )

print(top_15_global_cited)

write_csv(
  top_15_global_cited,
  "Figure_4_top_15_global_cited_documents.csv"
)

plot_global <- top_15_global_cited %>%
  mutate(
    figure_label = short_reference,
    figure_label = fct_reorder(figure_label, global_citations)
  )

p_global <- ggplot(plot_global, aes(x = figure_label, y = global_citations)) +
  geom_col(
    width = 0.65,
    fill = "#8FB9D9",
    color = "white",
    linewidth = 0.20
  ) +
  geom_text(

```

```

aes(label = global_citations),
hjust = -0.18,
size = 5.2,
color = "black",
fontface = "bold"
) +
coord_flip(clip = "off") +
scale_y_continuous(
  breaks = pretty_breaks(n = 6),
  expand = expansion(mult = c(0, 0.15))
) +
labs(
  title = NULL,
  subtitle = NULL,
  x = NULL,
  y = "Global citations"
) +
theme_minimal(base_size = 18) +
theme(
  axis.title.x = element_text(
    face = "bold",
    size = 21,
    color = "black",
    margin = margin(t = 14)
  ),
  axis.text.y = element_text(
    color = "black",
    size = 18
  ),
  axis.text.x = element_text(
    color = "black",
    size = 16
  ),
  panel.grid.minor = element_blank(),
  panel.grid.major.y = element_blank(),
  panel.grid.major.x = element_line(color = "grey86"),
  plot.margin = margin(15, 60, 15, 15)
)

```

```
print(p_global)
```

```

ggsave(
  filename = "Figure_4_Top_15_Most_Cited_Documents_Global_Citations.png",
  plot = p_global,
  width = 11,
  height = 8.5,
  units = "in",
  dpi = 600,
  limitsize = FALSE
)

```

)

```
ggsave(  
  filename = "Figure_4_Top_15_Most_Cited_Documents_Global_Citations.tiff",  
  plot = p_global,  
  width = 11,  
  height = 8.5,  
  units = "in",  
  dpi = 600,  
  compression = "lzw",  
  limitsize = FALSE  
)
```

06_citations_per_year.R

```
library(readr)
library(dplyr)
library(stringr)
library(ggplot2)
library(forcats)
library(scales)

M <- read_csv("scopus_wos_deduplicated.csv", show_col_types = FALSE)

required_cols <- c("title", "year", "source_title", "cited_by")
missing_cols <- required_cols[!required_cols %in% names(M)]

if (length(missing_cols) > 0) {
  cat("Missing required columns:\n")
  print(missing_cols)
  stop("Some required columns are missing. Check names(M).")
}

has_authors <- "authors" %in% names(M)
has_doi <- "doi" %in% names(M)

current_year <- 2026

cited_docs <- M %>%
  transmute(
    title = str_squish(as.character(title)),
    authors = if (has_authors) str_squish(as.character(authors)) else NA_character_,
    year = as.numeric(year),
    source_title = str_squish(as.character(source_title)),
    doi = if (has_doi) str_to_lower(str_squish(as.character(doi))) else NA_character_,
    global_citations = as.numeric(cited_by)
  ) %>%
  mutate(
    global_citations = ifelse(is.na(global_citations), 0, global_citations),

    years_since_publication = pmax(current_year - year + 1, 1),
    citations_per_year = global_citations / years_since_publication,

    first_author = case_when(
      is.na(authors) | authors == "" ~ "Unknown",
      str_detect(authors, ";") ~ str_squish(str_split_fixed(authors, ";", 2)[, 1]),
      str_detect(authors, ",") ~ str_squish(str_split_fixed(authors, ",", 2)[, 1]),
      TRUE ~ str_squish(str_split_fixed(authors, " ", 2)[, 1])
    ),

    short_reference = paste0(first_author, " (", year, ")")
  ) %>%
  filter(
```

```

  !is.na(title),
  title != "",
  !is.na(year),
  global_citations > 0
)

write_csv(
  cited_docs,
  "documents_with_global_citations_and_citations_per_year.csv"
)

top_15_cpy <- cited_docs %>%
  arrange(desc(citations_per_year), desc(global_citations)) %>%
  slice_head(n = 15) %>%
  mutate(
    rank_cpy = row_number(),
    citations_per_year = round(citations_per_year, 2)
  ) %>%
  select(
    rank_cpy,
    short_reference,
    year,
    title,
    source_title,
    global_citations,
    citations_per_year,
    doi
  )

print(top_15_cpy)

write_csv(
  top_15_cpy,
  "Figure_5_top_15_documents_by_citations_per_year.csv"
)

plot_cpy <- top_15_cpy %>%
  mutate(
    figure_label = short_reference,
    figure_label = fct_reorder(figure_label, citations_per_year)
  )

p_cpy <- ggplot(plot_cpy, aes(x = figure_label, y = citations_per_year)) +
  geom_col(
    width = 0.65,
    fill = "#8FB9D9",
    color = "white",
    linewidth = 0.20
  ) +

```

```

geom_text(
  aes(label = round(citations_per_year, 1)),
  hjust = -0.18,
  size = 5.2,
  color = "black",
  fontface = "bold"
) +
coord_flip(clip = "off") +
scale_y_continuous(
  breaks = pretty_breaks(n = 6),
  expand = expansion(mult = c(0, 0.18))
) +
labs(
  title = NULL,
  subtitle = NULL,
  x = NULL,
  y = "Citations per year"
) +
theme_minimal(base_size = 18) +
theme(
  axis.title.x = element_text(
    face = "bold",
    size = 21,
    color = "black",
    margin = margin(t = 14)
  ),
  axis.text.y = element_text(
    color = "black",
    size = 18
  ),
  axis.text.x = element_text(
    color = "black",
    size = 16
  ),
  panel.grid.minor = element_blank(),
  panel.grid.major.y = element_blank(),
  panel.grid.major.x = element_line(color = "grey86"),
  plot.margin = margin(15, 70, 15, 15)
)

```

```
print(p_cpy)
```

```

ggsave(
  filename = "Figure_5_Top_15_Documents_By_Citations_Per_Year.png",
  plot = p_cpy,
  width = 11,
  height = 8.5,
  units = "in",
  dpi = 600,

```

```
limitsize = FALSE
```

```
)
```

```
ggsave(
```

```
  filename = "Figure_5_Top_15_Documents_By_Citations_Per_Year.tiff",
```

```
  plot = p_cpy,
```

```
  width = 11,
```

```
  height = 8.5,
```

```
  units = "in",
```

```
  dpi = 600,
```

```
  compression = "lzw",
```

```
  limitsize = FALSE
```

```
)
```